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Micronutrient Deficiency among Pregnant Women: Evidence from Women and Infants Integrated Interventions for Growth Study (WINGS)

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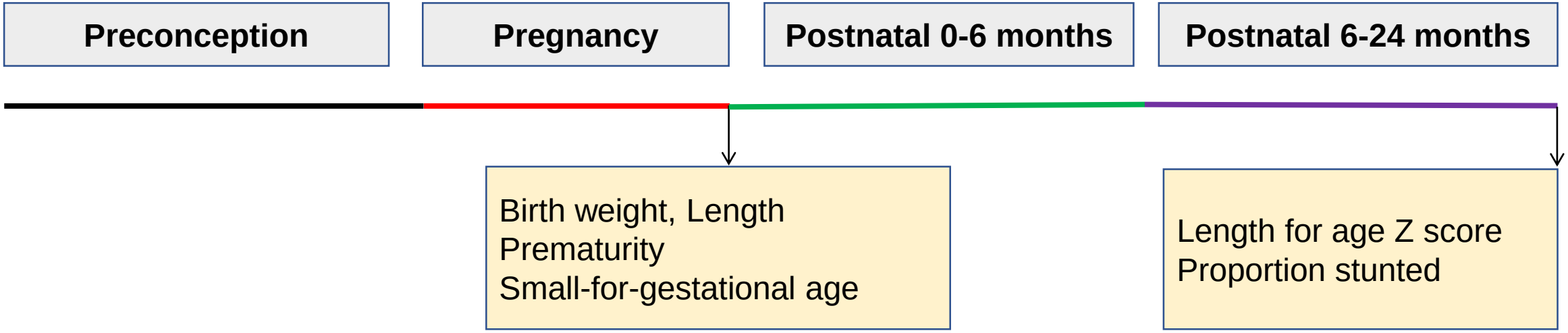
WINGS Study Design

Individually randomized, factorial design; non-pregnant 18-30 yr old married women living with husband, with no or one child and wish to have a child
Setting: Low and middle income neighbourhoods of Delhi, India



Intervention Package: Health, Nutrition, Water Sanitation & Hygiene, Care and Support

4 Groups: **Group A** (Preconception + Pregnancy and Early Childhood); **Group B** (Only Pre- and Peri-conception); **Group C** (Only Pregnancy and Early childhood); **Group D** (Routine Care)

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Interventions

	Health	Nutrition	Psychosocial Care	WASH
Preconception	Screen & treat medical conditions	Provide IFA & MMN, Provide egg/milk if BMI <21, Screen & treat malnutrition, anemia	Promote positive thinking & problem-solving skills	Promote personal, menstrual and hand hygiene
Pregnancy	≥8 antenatal contacts, screen & treat GHT/pre-eclampsia, GDM, hypothyroidism, UTI, RTI, calcium supplement	Provide IFA & MMN, Locally-prepared snacks (200/400 kcal in 2nd/3rd trimester), Provide milk (180 ml daily), Provide one additional hot cooked meal (500 kcal) if BMI<18.5 or inadequate gestational weight gain	Promote positive thinking & problem-solving skills	Provide water filters, soap, hand washing station, disinfectant
Early childhood	Empower family to identify danger signs and seek care early	0-6 mo: lactation support for early and exclusive BF 6-24 mo: promote timely CF & continued BF Provide supplementary food (125/250 kcal 6-11/12-23 mo) Double supplement if inadequate weight gain	Promote early child play and responsive care	Provide play mat and potty
Postnatal 6 mo: mother	Facilitate postnatal visit at 6 wk	Provide IFA, MMN, calcium & Vit D, locally-prepared snacks (600 kcal), provide milk (180 ml daily)	Promote positive thinking & problem-solving skills	Same as in pregnancy

Control group: Only weekly IFA supplementation as part of national program; routine antenatal and postnatal care

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Composition of Multiple Micronutrient (Riconia Silver)

Nutrients	WHO-RDA	RICONIA Silver (per tablet)	Riconia % RDA WHO
Vitamin A (mcg/d)	850	750	88%
Vitamin D (mcg/d)	5	5	100%
Vitamin K (mcg/d)	55	10	18%
Vitamin C (mg/d)	70	50	71%
Vitamin E (mg a-TE/d)	7.5	10	133%
Thiamine (mg/d)	1.5	2	133%
Riboflavin (mg/d)	1.6	3	188%
Niacin (mg/d)	17	26	153%
Pyridoxine (mg/d)	2	1.5	75%
Pantothenate (mg/d)	7	5	71%
Folate (mcg/d)	500	300	60%
Vitamin B12 (mcg/d)	2.8	1	36%
Biotin (mcg/d)	35	30	86%
Iron (mg/d)	15	9	60%
Calcium (mg/d)	1000	162	16%

Nutrients	WHO-RDA	RICONIA Silver (per tablet)	Riconia % RDA WHO
Zinc (mg/d)	9.5	15	158%
Iodine (mcg/d)	200	150	75%
Selenium (mcg/d)	35	25	71%
Magnesium (mg/d)	270	100	37%
Copper (mg/d)		2	
Manganese (mg/d)		2.5	
Phosphorus (mg/d)		125	
Potassium (mg/d)		40	
Phosphorus (mg/d)		125	
Chloride (mg/d)		36.3	
Chromium (mcg/d)		25	
Molybdenum (mcg/d)		25	
Nickel (mcg/d)		5	
Silicon (mcg/d)		10	
Vanadium (mcg/d)		10	

All pregnant women received one iron-folic acid tablet (elemental iron 100 mg + folic acid 500 μg) everyday

Riconia Silver, Sun Pharmaceutical Industries Ltd, commercially available
Micronutrients highlighted in yellow not present in UNIMMAP

Micronutrient Status in Pregnant Women: 11-13 Weeks of Gestation*

	n=397
Vitamin A	
Mean (SD)	0.42 (0.18)
Proportion deficient (<0.196 mg/L)	24 (6.1%)
Vitamin D-3 (ng/mL)	
Mean (SD)	9.1 (8.3)
Proportion deficient (<12 ng/mL)	314 (79.1)
Vitamin B12 (pg/mL)	
Mean (SD)	252.0 (195.6)
Proportion deficient (<203 pg/mL)	202 (50.9)
Zinc (µg/dL)	
Mean (SD)	102.5 (74.8)
Proportion deficient (<66 µg/dL)	140 (35.3)
Ferritin (ng/mL)	
Mean (SD)	34.1 (54.9)
Proportion deficient (CRP ≤5 mg/L & Ferritin <15 ng/mL or CRP > 5 mg/L & Ferritin < 70 ng/mL)	207 (52.1)
Folic Acid (ng/mL)	
Mean (SD)	10.5 (6.1)
Proportion deficient (<4 ng/mL)	58 (14.6)
Selenium (ug/L)	
Mean (SD)	124.0 (17.6)
Proportion deficient (<85 ug/L)	5 (1.4)

*Routine care group during preconception period
Laboratory: Dr. Dangs Lab, New Delhi

Micronutrient Status at 35-37 Weeks of Gestation

	Pre-conception and pregnancy (Group A) n=248	Pre-conception only (Group B) n=247	Pregnancy only (Group C) n=250	Routine care (Group D) n=250
Vitamin A , mg/L				
Mean (SD)	0.48 (0.21)	0.39 (0.18)	0.48 (0.22)	0.36 (0.20)
Proportion deficient (<0.196 mg/L)	12 (5.8)	23 (11.2)	13 (6.0)	44 (20.7)
Vitamin D-3 (ng/mL)				
Mean (SD)	17.0 (9.7)	8.8 (6.5)	18.7 (11.5)	9.8 (7.7)
Proportion deficient (<12 ng/mL)	89 (35.9)	196 (79.4)	89 (35.6)	189 (75.6)
Vitamin B12 (pg/mL)				
Mean (SD)	236.7 (153.6)	212.6 (175.5)	248.9 (232.0)	218.4 (214.5)
Proportion deficient (<203 pg/mL)	127 (51.2)	158 (64.0)	140 (56.0)	158 (63.2)
Zinc (µg/dL)				
Mean (SD)	92.9 (84.0)	96.2 (67.5)	90.1 (67.1)	88.4 (72.0)
Proportion deficient (<66 µg/dL)	126 (50.8)	100 (40.5)	114 (45.8)	124 (49.6)
Ferritin (ng/mL)				
Mean (SD)	46.0 (52.5)	30.5 (35.2)	50.1 (65.9)	29.8 (33.5)
Proportion deficient (CRP ≤5 mg/L & Ferritin <15 ng/mL or CRP >5 mg/L & Ferritin <70 ng/mL)	90 (36.3)	121 (49.0)	94 (37.6)	125 (50.0)
Folic Acid (ng/mL)				
Mean (SD)	11.8 (5.9)	8.1 (5.8)	12.4 (5.9)	7.8 (5.6)
Proportion deficient (<4 ng/mL)	38 (15.3)	82 (33.2)	29 (11.6)	81 (32.4)
Selenium (ug/L)				
Mean (SD)	123.8 (18.5)	121.9 (18.1)	126.5 (17.0)	122.9 (18.3)
Proportion deficient (<85 ug/L)	1 (0.4)	4 (1.7)	1 (0.4)	3 (1.3)

Laboratory: Dr. Dangs Lab, New Delhi

Intervention Effect on Micronutrient Status at 35-37 Weeks of Gestation

Outcomes	Comparisons					
	Preconception interventions vs no preconception interventions (A+B v C+D)		Pregnancy interventions vs no pregnancy interventions (A+C v B+D)		Preconception and pregnancy interventions vs no preconception and pregnancy interventions (A v D)	
	IRR (98.3%)	ARR (%) or MD (98.3%)	IRR (98.3%)	ARR (%) or MD (98.3%)	IRR (98.3%)	ARR (%) or MD (98.3%)
Vitamin A, mg/L		0.01 (-0.02, 0.05)		0.10 (0.07, 0.14)		0.11 (0.07, 0.16)
Vitamin A deficiency	0.64 (0.38, 1.07)	-4.23 (-9.69, 1.23)	0.37 (0.21, 0.64)	-10.11 (-15.54, -4.69)	0.27 (0.12, 0.59)	-13.77 (-22.15, -5.39)
Vitamin D, ng/mL		-1.31 (-2.83, 0.21)		8.50 (7.12, 9.88)		7.14 (5.25, 9.02)
Vitamin D deficiency	1.03 (0.84, 1.26)	1.88 (-9.51, 13.27)	0.46 (0.37, 0.57)	-42.12 (-53.52, -30.72)	0.48 (0.35, 0.65)	-39.82 (-55.81, -23.82)
Vitamin B12, pg/mL		-9.33 (-39.16, 20.51)		27.14 (-2.63, 56.90)		16.79 (-23.40, 56.98)
Vitamin B12 deficiency	0.97 (0.79, 1.18)	-2.25 (-13.82, 9.32)	0.84 (0.69, 1.03)	-10.21 (-21.77, 1.35)	0.82 (0.61, 1.08)	-12.42 (-28.52, 3.69)
Zinc, µg/dL		5.09 (-5.97, 16.15)		-1.07 (-12.13, 9.99)		4.08 (-12.69, 20.84)
Zinc deficiency	0.96 (0.77, 1.20)	-2.17 (-12.57, 8.22)	1.07 (0.85, 1.33)	3.02 (-7.35, 13.38)	1.02 (0.75, 1.38)	0.62 (-14.63, 15.87)

Adjusted for place of birth, below poverty line card, woman's height, woman BMI as potential confounders and twins for clustering within a household. IRR: Incidence rate ratio, ARR: Absolute risk reduction, MD: Mean difference
Laboratory: Dr. Dangs Lab, New Delhi

Intervention Effect on Micronutrient Status at 35-37 Weeks of Gestation

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	Preconception interventions vs no preconception interventions (A+B v C+D)		Pregnancy interventions vs no pregnancy interventions (A+C v B+D)		Preconception and pregnancy interventions vs no preconception and pregnancy interventions (A v D)	
	IRR (98.3%)	ARR (%) or MD (98.3%)	IRR (98.3%)	ARR (%) or MD (98.3%)	IRR (98.3%)	ARR (%) or MD (98.3%)
Ferritin, ng/mL		-1.95 (-9.43, 5.53)		17.92 (10.57, 25.28)		16.01 (6.57, 25.44)
Ferritin deficiency	0.98 (0.78, 1.23)	-0.89 (-10.86, 9.08)	0.75 (0.60, 0.95)	-12.26 (-22.21, -2.30)	0.74 (0.53, 1.03)	-13.15 (-27.29, 1.00)
Folic Acid, ng/mL		-0.15 (-1.09, 0.78)		4.17 (3.29, 5.05)		3.98 (2.74, 5.23)
Folic acid deficiency	1.10 (0.80, 1.51)	2.04 (-5.26, 9.34)	0.40 (0.29, 0.57)	-19.37 (-26.63, -12.10)	0.46 (0.29, 0.74)	-17.39 (-27.75, -7.03)
Selenium, ug/L		-1.80 (-4.63, 1.02)		2.69 (-0.14, 5.51)		1.00 (-3.12, 5.12)
Selenium deficiency	1.31 (0.26, 6.66)	0.22 (-1.32, 1.76)	0.29 (0.04, 1.98)	-1.09 (-2.63, 0.46)	0.30 (0.02, 4.75)	-0.87 (-2.92, 1.19)

Adjusted for place of birth, below poverty line card, woman's height, woman BMI as potential confounders and twins for clustering within a household. IRR: Incidence rate ratio, ARR: Absolute risk reduction, MD: Mean difference
Laboratory: Dr. Dangs Lab, New Delhi

Summary

Pregnancy interventions improved vitamin A, D, iron and folic acid status during pregnancy

Effect of interventions delivered during both preconception and pregnancy periods are similar

Thank you

Micronutrient Status in Lactating Women at 6 Months Postnatal Period

	Pre-conception and pregnancy (Group A) n=275	Pre-conception only (Group B) n=258	Pregnancy only (Group C) n=189	Routine care (Group D) n=193
Vitamin A , mg/L Mean (SD) Proportion deficient (<0.196 mg/L)	0.5 (0.2) 22 (8.1)	0.4 (0.2) 25 (9.7)	0.5 (0.2) 10 (5.4)	0.4 (0.2) 15 (7.8)
Vitamin D-3 (ng/mL) Mean (SD) Proportion deficient (<12 ng/mL)	10.7 (6.3) 191 (69.5)	9.3 (6.7) 211 (81.8)	13.6 (7.0) 96 (50.8)	9.0 (5.1) 151 (78.2)
Vitamin B12 (pg/mL) Mean (SD) Proportion deficient (<203 pg/mL)	278.5 (154.9) 86 (31.3)	304.1 (151.9) 65 (25.2)	307.5 (138.6) 32 (16.9)	261.2 (110.1) 69 (35.8)
Zinc (µg/dL) Mean (SD) Proportion deficient (<66 µg/dL)	82.1 (55.9) 129 (46.9)	88.0 (60.4) 116 (45.0)	81.3 (49.4) 85 (45.0)	80.2 (52.3) 96 (49.7)
Ferritin (ng/mL) Mean (SD) Proportion deficient (CRP ≤5 mg/L & Ferritin <15 ng/mL or CRP >5 mg/L & Ferritin <70 ng/mL)	54.6 (48.9) 103 (37.5)	35.6 (31.8) 143(55.4)	62.5 (51.5) 54 (28.6)	36.6 (35.3) 107 (55.4)
Folic Acid (ng/mL) Mean (SD) Proportion deficient (<4 ng/mL)	5.5 (4.2) 135 (49.1)	4.3 (2.6) 148 (57.4)	6.8 (4.7) 73 (38.6)	4.2 (2.6) 117 (60.6)
Selenium (ug/L) Mean (SD) Proportion deficient (<85 ug/L)	117.8 (18.5) 4 (2.2)	118.9 (17.0) 4 (2.0)	115.9 (19.6) 7 (5.8)	118.1 (15.9) 2 (1.9)

Micronutrient Status in Lactating Women at 6 Months Postnatal Period

Outcomes	Comparisons					
	Preconception interventions vs no preconception interventions (A+B v C+D)		Pregnancy interventions vs no pregnancy interventions (A+C v B+D)		Preconception and pregnancy interventions vs no preconception and pregnancy interventions (A v D)	
	IRR (98.3%)	ARR (%) or MD (98.3%)	IRR (98.3%)	ARR (%) or MD (98.3%)	IRR (98.3%)	ARR (%) or MD (98.3%)
Vitamin A, mg/L		-0.04 (-0.04, 0.03)		0.08 (0.04, 0.11)		0.08 (0.03, 0.13)
Vitamin A deficiency	1.34 (0.76, 2.38)	1.23 (-3.04, 5.50)	0.66 (0.37, 1.20)	-3.27 (-7.29, 0.75)	0.93 (0.42, 2.09)	-1.19 (-7.99, 5.62)
Vitamin D, ng/mL		-1.24 (-2.28, -0.19)		2.98 (1.94, 4.03)		1.92 (0.55, 3.29)
Vitamin D deficiency	1.17 (1.05, 1.30)	10.86 (3.48, 18.25)	0.75 (0.67, 0.84)	-20.28 (-27.74, -12.81)	0.86 (0.75, 0.99)	-11.06 (-21.42, -0.69)
Vitamin B12, pg/mL		8.80 (-13.37, 30.97)		12.74 (-9.71, 35.19)		23.78 (-5.94, 53.49)
Vitamin B12 deficiency	1.04 (0.80, 1.35)	-0.98 (-8.55, 6.59)	0.76 (0.58, 0.99)	-6.46 (-13.68, 0.75)	0.80 (0.58, 1.10)	-8.20 (-20.01, 3.62)
Zinc, µg/dL		5.10 (-3.26, 13.45)		1.07 (-7.65, 9.78)		4.54 (-7.11, 16.19)
Zinc deficiency	0.97 (0.82, 1.15)	-1.52 (-9.67, 6.62)	0.95 (0.79, 1.14)	-2.51 (-10.93, 5.91)	0.92 (0.72, 1.18)	-4.40 (-16.55, 7.76)

Adjusted for place of birth, below poverty line card, woman's height, woman BMI as potential confounders and twins for clustering within a household. IRR: Incidence rate ratio, ARR: Absolute risk reduction, MD: Mean difference

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	Preconception interventions vs no preconception interventions (A+B v C+D)		Pregnancy interventions vs no pregnancy interventions (A+C v B+D)		Preconception and pregnancy interventions vs no preconception and pregnancy interventions (A v D)	
	IRR (98.3%)	ARR (%) or MD (98.3%)	IRR (98.3%)	ARR (%) or MD (98.3%)	IRR (98.3%)	ARR (%) or MD (98.3%)
Ferritin, ng/mL		-4.50 (-11.60, 2.60)		21.58 (14.65, 28.52)		17.06 (7.27, 26.85)
Ferritin deficiency	1.13 (0.94, 1.35)	5.61 (-2.40, 13.62)	0.63 (0.51, 0.76)	-20.22 (-28.35, -12.08)	0.71 (0.55, 0.93)	-15.89 (-28.23, -3.55)
Folic Acid, ng/mL		-0.52 (-1.13, -0.08)		1.73 (1.13, 2.34)		1.32 (0.50, 2.14)
Folic acid deficiency	1.08 (0.92, 1.26)	3.98 (-4.15, 12.10)	0.75 (0.64, 0.89)	-14.27 (-22.79, -5.74)	0.83 (0.66, 1.03)	-10.66 (-23.62, 2.30)
Selenium, ug/L		1.27 (-2.30, 4.85)		-155 (-5.21, 2.11)		-0.86 (-5.97, 4.25)
Selenium deficiency	0.56 (0.17, 1.82)	-1.92 (-5.49, 1.65)	1.79 (0.51, 6.24)	1.71 (-1.48, 4.90)	1.35 (0.21, 8.77)	0.30 (-3.86, 4.45)

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